

Smart Food Delivery System - Full Project Roadmap

1. Understanding the Goal

Build a real-time delivery system using advanced tree data structures with features like nearest partner assignment, priority handling, batching, and reassignment.

2. Tech Stack Recommendation

- 1 Frontend: React / HTML-CSS-JS
- 2 Backend: Python (Flask/FastAPI) or Node.js
- 3 Data Structures: Custom implementation (KD-Tree, Treap)
- 4 Database: Optional (SQLite / MongoDB)
- 5 Simulation: In-memory system for faster testing

3. Phase 1 - Basic System Setup

- 1 Define Order and Partner classes
- 2 Create APIs: Add partner, place order
- 3 Basic assignment logic (brute force)

4. Phase 2 - KD-Tree Implementation

- 1 Build KD-Tree for partner locations
- 2 Implement nearest neighbor search
- 3 Replace brute force assignment with KD-Tree

5. Phase 3 - Priority Orders (Heap)

- 1 Implement priority queue (heap)
- 2 Process high-priority orders first
- 3 Maintain two queues (normal + priority)

6. Phase 4 - Order Batching

- 1 Group nearby orders using spatial distance
- 2 Assign batch to single partner
- 3 Limit batch size (2-5 orders)

7. Phase 5 - Treap for Dynamic Reassignment

- 1 Implement Treap (BST + Heap)
- 2 Store partners based on availability
- 3 Reassign when priority order arrives
- 4 Handle conflict resolution

8. Phase 6 - Full Workflow Integration

- 1 Order placed → classified
- 2 Find nearest partner
- 3 Check batching
- 4 Handle reassignment
- 5 Update system state

9. Phase 7 - UI / Simulation

- 1 Map view (optional)
- 2 Show partners & orders
- 3 Real-time assignment logs
- 4 Buttons for testing scenarios

10. Phase 8 - Edge Case Handling

- 1 No available partners
- 2 Multiple priority orders
- 3 Reassignment loops
- 4 System overload

11. Phase 9 - Optimization

- 1 Reduce KD-Tree search time
- 2 Efficient rebalancing

- 3 Improve batching logic

12. Phase 10 - Final Polish

- 1 Add logs and metrics
- 2 Test performance
- 3 Prepare demo scenarios
- 4 Document code

13. Bonus Features

- 1 Route optimization
- 2 Live tracking
- 3 ML-based demand prediction